EDUKRIPA

CLASS - 10 CBSE **SCIENCE TEST - 2** **Marks: 40**

(Chemical Reactions & Equations, Nutrition and Light)

1. Identify '*x*', '*y*', and '*z*' in the following balanced reaction:

   $xAl(s) + yO_2(g) \rightarrow z\ Al_2O_3\ (s)$

   (a) 4, 3, 2 (b) 2, 1, 1 (c) 4, 2, 2 (d) 2, 3, 2

2. Which of the following structures is involved in gaseous exchange in woody stem of a plant as shown in the figure?

   (a) Stomata (b) Guard cell (c) Lenticel (d) Epidermis

3. Exposure of silver chloride to sunlight for a long duration turns grey due to:

   Which among the following statement(s) is(are) true?

   (i) the formation of silver by decomposition of silver chloride.

   (ii) sublimation of silver chloride.

   (iii) decomposition of chlorine gas from silver chloride.

   (iv) oxidation of silver chloride.

   (a) Only 1 (b) 1 and 3 (c) 2 and 3 (d) Only 4

4. **Assertion :** Photosynthesis is considered as an endothermic reaction.

   Reason : Energy gets released in the process of photosynthesis.

   (a) Both Assertion and Reason are True and Reason is the correct explanation of the Assertion.

   (b) Both Assertion and Reason are True but Reason is not the Correct explanation of the Assertion.

   (c) Assertion is True but the Reason is False.

   (d) Both Assertion and Reason are False.

5. **Assertion :** Our body maintains blood sugar level.

   Reason : Pancreas secretes insulin which helps to regulate blood sugar levels in the body.

   (a) Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of (A).

   (b) Both assertion (A) and reason (R) are true but (R) is not the correct explanation of assertion (A).

   (c) Assertion (A) is true but reason (R) is false.

   (d) Assertion (A) is false but reason (R) is true.

6. **Assertion :** Artificial kidney is a device used to remove nitrogenous waste products from the blood through dialysis.

   Reason : Reabsorption does not occur in artificial kidney.

(a) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(b) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.

(c) Assertion is true but Reason is false.

(d) Assertion is false but Reason is true.

7. Process of conversion of light energy to chemical energy and splitting of water molecules into hydrogen and oxygen in plants is known as

(a) Photosynthesis (b) Photoperiodism (c) Plant nutrition (d) Plant hormone functions

8. What happens when water is added to quick lime? Write chemical equation. [2]

9. Why and how does water enter continuously into the root xylem ? [2]

10. A ray of light is incident on a plane mirror at an angle of 60° with the mirror surface. Calculate: [2]

i. The angle of incidence and the angle of reflection.

ii. The angle between the incident ray and the reflected ray.

11. State the function of Bowman's Capsule and glomerulus. [2]

12. Write one function each of the salivary glands, liver and pancreas. [3]

13. Differentiate between blood and lymph. [3]

14. A person walks 3 m towards a plane mirror from an initial distance of 8 m. [3]

a) What is his final distance from the mirror?
b) What is the distance between him and his image now?
c) By how much has the distance between the person and his image decreased?

15. How is O_2 and CO_2 transported in human beings? [3]

16. Describe the structure and function of the basic filtering unit of kidney with diagram. Also, explain the formation of urine. [5]

17. Read the following passage carefully and answer the questions that follow: [4]

A student carried out three different experiments in the laboratory.

- In Experiment A, a clean iron nail was dipped in copper sulphate solution for some time. The blue colour of the solution gradually faded and a reddish-brown coating appeared on the iron nail.
- In Experiment B, lead nitrate solution was mixed with potassium iodide solution. A bright yellow precipitate was formed immediately.
- In Experiment C, silver chloride was exposed to sunlight for a few hours. The white solid gradually turned grey.

The reactions are represented as:

a. $Fe + CuSO_4 \rightarrow FeSO_4 + Cu$

b. $Pb(NO_3)_2 + 2KI \rightarrow PbI_2 + 2KNO_3$

c. $2AgCl \rightarrow 2Ag + Cl_2$

i. What evidence suggests that a chemical reaction has taken place in Experiment A?

ii. Name the yellow precipitate formed in Experiment B. What type of reaction is it?

iii. Why does silver chloride turn grey on exposure to sunlight?

iv. Predict what will happen if a copper strip is dipped in iron sulphate solution. Give a reason.

18. Questions are based on the two table given below. Study these tables related to blood pressure level and answer the question that follow : [4]

Table A

BLOOD PRESSURE	**SYSTOLIC mm Hg**	**DIASTOLIC mm Hg**
Normal	**120**	**80**
Elevated	120–129	Less than 80
High B.P Stage 1	130–139	80–90
High B.P Stage 2	140 or higher	90 or higher
Hypertensive crisis	Higher than 180	Higher than 120

Table B

Time of Measurement	**Blood Pressure**	
	Patient–X	**Patient–Y**
Morning	75–115	85–125
Afternoon	79–122	80–120
evening	82-132	75-110

(i) In the table B, at which time patent–Y have ideal normal blood pressure ?

(ii) Identify the patient, which have hypertension stage-1 blood pressure ?

(iii) Which Diet is the best for high blood pressure patient ?

OR

(iv) What is the ideal blood pressure measurement of a human ?